An improved upper bound for binary covering codes of length 20 and covering radius 3

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Abstract

Let $K(n, R)$ denote the minimum number of words in a binary code of length n with covering radius R. We show $K(20, 3) \le 1984$, improving the bound $K(20, 3) \le 2048$ recorded in Kéri's standard tables of covering codes. The code is the amalgamated direct sum of the $[7, 4]$ Hamming code with a new *normal* binary code of length 14, covering radius 2 and 248 words (a union of 62 cosets of a $[14, 2, 3]$ linear code). The covering property is certified in Lean 4 with Mathlib, using only the standard axioms. We also report, as negative results, extensive but unsuccessful computer searches aimed at the football pool problem $K_3(n, 1)$ for $n = 6, 7, 8$.

1 Introduction

A binary code $C \subseteq \mathbb{F}_2^n$ has *covering radius* R if every word of $\mathbb{F}_2^n$ is within Hamming distance R of some codeword. The quantity $K(n, R)$, the minimum size of such a code, has been studied since the 1960s; see the monograph of Cohen, Honkala, Litsyn and Lobstein [1] and the football pool problem $K_3(n, 1)$ for its ternary analogue [5, 6]. The best known bounds for $n \le 33$, $R \le 10$ are collected in G. Kéri's tables [4], which have been the standard reference since the 2000s (the binary table we consulted is dated October 2009; its update log ends in November 2011). In these tables

$$892 \;\le\; K(20, 3) \;\le\; 2048,$$

where the upper bound carries no reference mark, i.e. it is classified as "trivial" (inherited from neighbouring entries).

Result. We prove $K(20, 3) \le 1984$ (Theorem 4.1). The construction is classical — the amalgamated direct sum (ADS) of Graham and Sloane [3] and Cohen, Lobstein and Sloane [2] — and the observation behind it is simple: the ADS of the $[7, 4]$ Hamming code with a *normal* code of length 14 and covering radius 2 having M words has length 20, covering radius 3 and $8M$ words. With $M = 256 = 2 \cdot 128$ (twice the tabulated bound for $K(13, 2)$) this gives exactly the recorded value 2048, which suggests how the entry may have arisen; the table entry $K(14, 2) \le 248$ (credited in [4] to Bertolo, Di Pasquale and Santisi, 2006) would give 1984, provided some 248-word code is normal — a property that is not automatic and that, as far as we can tell, was never established

or used. We construct such a normal code directly (Section 3) and verify the whole argument in Lean.

To our knowledge, as of October 14, 2026, no upper bound below 2048 for $K(20, 3)$ has been published; we searched the update log of [4] and recent work on covering codes [7, 8, 9], which concerns lower bounds, alphabets $q \geq 5$ and linear codes respectively. A systematic check of all binary and mixed binary/ternary table entries under direct sums, puncturing, extension and ADS (assuming normality) found no other entry that should have improved in this way.

This is a modest improvement of one table entry. The work started as an attempt at the football pool problem (Section 6); we report those negative results as well.

2 Definitions

For $x, y \in \mathbb{F}_2^n$ let $d(x, y)$ be the Hamming distance and $d(x, C) = \min_{c \in C} d(x, c)$. For a code C and $s \in \{0, 1\}$ write $C_i^s = \{c \in C : c_i = s\}$.

Definition 2.1 (Normality)**.** A code C of covering radius R is *normal with respect to coordinate* i if $d(x, C_i^0) + d(x, C_i^1) \leq 2R + 1$ for all $x \in \mathbb{F}_2^n$.

Definition 2.2 (Amalgamated direct sum)**.** Let $A \subseteq \mathbb{F}_2^{n_A}$ and $B \subseteq \mathbb{F}_2^{n_B}$. The ADS (amalgamated on the last coordinate of A and the first coordinate of B) is

$$A \dot{\oplus} B = \{(a_1, \ldots, a_{n_A-1}, s, b_2, \ldots, b_{n_B}) : (a_1, \ldots, a_{n_A-1}, s) \in A,\ (s, b_2, \ldots, b_{n_B}) \in B\} \subseteq \mathbb{F}_2^{n_A+n_B-1}.$$

Clearly $|A \dot{\oplus} B| = |A^0||B^0| + |A^1||B^1|$, where A^s (resp. B^s) is taken with respect to the amalgamated coordinate.

Proposition 2.3 ([3, 2])**.** *If A has covering radius R_A and is normal w.r.t. its last coordinate, and B has covering radius R_B and is normal w.r.t. its first coordinate, then $A \dot{\oplus} B$ has covering radius at most $R_A + R_B$.*

Proof. Write $x = (\alpha, t, \beta)$ with $\alpha \in \mathbb{F}_2^{n_A-1}$, $t \in \mathbb{F}_2$, $\beta \in \mathbb{F}_2^{n_B-1}$, and for $s \in \{0, 1\}$ choose $a^s \in A^s$, $b^s \in B^s$ nearest to (α, t) and (t, β). The word c^s obtained by amalgamating a^s and b^s lies in $A \dot{\oplus} B$ and $d(x, c^s) = d((\alpha, t), a^s) + d((t, \beta), b^s) - [t \neq s]$. Summing over s and using normality, $d(x, c^0) + d(x, c^1) \leq (2R_A + 1) + (2R_B + 1) - 1$, so $\min_s d(x, c^s) \leq R_A + R_B$. □

3 A normal code of length 14 and covering radius 2 with 248 words

Number the coordinates $0, \ldots, 13$ and identify a word with the integer $\sum_i x_i 2^i$. Let $H = [I_{12} \mid h_{12}\ h_{13}]$ be the 12×14 binary matrix whose columns are $e_0, \ldots, e_{11}$, $h_{12} = e_1 + e_2$ and $h_{13} = e_0 + e_1 + e_2$ (as integers: $1, 2, 4, \ldots, 2048, 6, 7$). Its kernel is the $[14, 2, 3]$ code spanned by $e_1 + e_2 + e_{12}$ and $e_0 + e_1 + e_2 + e_{13}$. Let $S \subseteq \mathbb{F}_2^{12}$ be the following set of 62 syndromes (integers, bit j = row j):

4, 74, 188, 231, 322, 377, 401, 428, 553, 599, 655, 738, 783, 820, 986, 993, 1055, 1138, 1194, 1236, 1313, 1468, 1487, 1562, 1601, 1719, 1785, 1879, 1900, 1922, 2097, 2114, 2220, 2271, 2330, 2343, 2505, 2546, 2575, 2684, 2706, 2791, 2897, 2922, 2948, 3007, 3183, 3201, 3258, 3338, 3346, 3420, 3556, 3575, 3620, 3665, 3788, 3903, 3905, 3993, 4015, 4082.

Define $B = \{x \in \mathbb{F}_2^{14} : Hx \in S\}$, a union of 62 cosets of $\ker H$, so $|B| = 248$.

Proposition 3.1. *B has covering radius 2, $|B^0| = |B^1| = 124$ with respect to coordinate 0, and B is normal with respect to coordinate 0: $d(y, B^0) + d(y, B^1) \le 5$ for all $y \in \mathbb{F}_2^{14}$.*

The proof is a finite computation over the 2^{14} words (it is part of the Lean certificate, Section 5; independently, a breadth-first computation of $d(\cdot, B^0)$ and $d(\cdot, B^1)$ gives $\max_y(d(y, B^0) + d(y, B^1)) = 5$). In fact B is normal with respect to every coordinate. The size 248 equals the best known upper bound for $K(14, 2)$ [4]; it does not improve it.

How B was found. Codes of the form $\{x : Hx \in S\}$ with $H = [I_k \mid h_k \ \cdots \ h_{n-1}]$ have covering radius $\le R$ iff S is an R-dominating set in the Cayley graph of $\mathbb{F}_2^k$ generated by the columns of H (the "matrix method"). For $n = 14$, $k = 12$ there are only 239 classes of the two extra columns up to coordinate permutations (given by the numbers of coordinates in which only h_{12}, only h_{13}, or both are 1). We ran a weighted local search for small 2-dominating sets on each class; the class $(0, 1, 2)$, i.e. $h_{12} = e_1 + e_2$, $h_{13} = e_0 + e_1 + e_2$, gave $|S| = 62$ within seconds (it is the only class reaching 62 in our runs; 194 classes reach 64), and normality was then checked.

4 The main theorem

Let A be the $[7, 4]$ Hamming code with parity-check columns $1, 2, \ldots, 7$ (coordinate j has column $j+1$): $A = \{0, 7, 25, 30, 42, 45, 51, 52, 75, 76, 82, 85, 97, 102, 120, 127\}$. It is perfect, $|A^0| = |A^1| = 8$ with respect to coordinate 6, and normal with respect to every coordinate (one checks $d(y, A^0) + d(y, A^1) \le 3$ for all $y \in \mathbb{F}_2^7$).

Theorem 4.1. *$C = A \dot{\oplus} B$, amalgamated on coordinate 6 of A and coordinate 0 of B, is a binary code of length 20 with 1984 words and covering radius 3. Hence $K(20, 3) \le 1984$.*

Proof. Length $7 + 14 - 1 = 20$; size $8 \cdot 124 + 8 \cdot 124 = 1984$; covering radius $\le 1 + 2$ by Theorems 2.3 and 3.1. (A word at distance exactly 3 exists, e.g. by the sphere bound.) In integer form, $C = \{(u \bmod 64) + 64v : u \in A,\ v \in B,\ \lfloor u/64 \rfloor = v \bmod 2\}$. □

Remark 4.2. Equivalently, $C = \{x \in \mathbb{F}_2^{20} : H_C x \in \{0\} \times S\}$, where H_C is the 15×20 matrix whose first three rows are the parity checks of A on coordinates $0, \ldots, 6$ and whose last twelve rows are H on coordinates $6, \ldots, 19$; thus C is a union of 62 cosets of a $[20, 5]$ linear code.

Remark 4.3. A brute-force check over all 2^{20} words confirms that the 1984 words are distinct and that every word is within distance 3 of the code. The data files accompany this note (`data/`).

5 Formal verification

The file `lean/Main.lean` (Lean 4.35.0, Mathlib) proves

```
theorem main : ∃ C : Finset (Fin 20 → Fin 2), C.card ≤ 1984 ∧
    ∀ x : Fin 20 → Fin 2, ∃ c ∈ C, hammingDist x c ≤ 3
```

and the companion statement `codeB` for B (length 14, ≤ 248 words, covering radius ≤ 2). Here `hammingDist` is Mathlib's Hamming distance (number of coordinates where two functions differ), so the statement is literally $K(20, 3) \le 1984$.

Structure of the proof. Words are encoded by natural numbers $a < 2^n$ through their binary digits (`dec n a`); the lemma `hamming_dec` proves that `hammingDist` of two decoded words equals a recursively defined digit-difference count `dist n a b`, and `dec_surj` (via Mathlib's `finFunctionFinEquiv`) shows that every word is decoded from some $a < 2^n$. The finite facts — normality of A over 128 words and normality of B over 2^{14} words, the latter with packed witness pairs checked by a tail-recursive Boolean function whose soundness lemma is proved — are established by `decide +kernel`, i.e. evaluated by the Lean kernel; no `native_decide` is used. The ADS covering argument of Theorem 2.3 is proved in Lean (lemmas `key` and `cover_nat`).

Axioms. `#print axioms` reports `[propext, Classical.choice, Quot.sound]` for both `main` and `codeB`. Checking took 105–151 s wall time and about 12 GB peak memory on one core.

Statement audit. `Fin 20 → Fin 2` is the set of binary words of length 20; `C.card ≤ 1984` together with the covering condition is exactly the definition of $K(20, 3) \leq 1984$. The code is not hidden behind a definition in the statement: the existential is discharged by an explicit list.

6 Negative results on the football pool problem

Our primary target was an upper bound in the football pool problem $K_3(n, 1)$, whose records $K_3(6, 1) \leq 73$, $K_3(7, 1) \leq 186$, $K_3(8, 1) \leq 486$ date from 1989 [5]. We did not improve any of them. For the record:

- $n = 7$: the matrix method with $k = 6$ (all six inequivalent choices of the seventh column) yields at best $3 \cdot 62 = 186$; with $|S| = 61$ our local search always left at least one vertex uncovered. The methods with $k = 5, 4$ gave at best 189 and 216. We also treated every conjugacy class of fixed-point-free automorphisms of order 3 of the Hamming graph $H(7, 3)$ (products of $c \in \{0, 1, 2\}$ coordinate 3-cycles with a symbol translation on $w \geq 1$ fixed coordinates; 12 classes): codes invariant under such an automorphism correspond to dominating sets of a 729-vertex quotient graph, and the best size found was again $3 \cdot 62 = 186$. Allowing a few non-invariant extra words ("almost invariant" codes: 60 or 61 orbits plus an exactly optimised set of extra words covering the remaining uncovered words) never went below 186. A weighted local search on the full space $\mathbb{F}_3^7$ reached only 192–193; codes invariant under random subgroups of the automorphism group (orbit local search and integer programming) also failed.
- $n = 8$: the matrix method reproduces 486 ($k = 6, |S| = 54$ and $k = 4, |S| = 6$) but nothing smaller.
- $n = 6$: 73-word codes are found in milliseconds, 72 never.

A sweep over the open entries of Kéri's binary, ternary, quaternary and mixed tables with full-space local search reproduced table values for small parameters and never improved one; the records reflect algebraic structure that unstructured search does not find.

7 Open questions

Any normal binary code of length 14, covering radius 2 and $M < 248$ words would give $K(20, 3) \leq 8M$ and at the same time improve $K(14, 2) \leq 248$. Our searches (all two-extra-column classes of

the matrix method with $k = 12$, one-extra-column classes with $k = 13$, and codes formed by 123 cosets of a $[14, 1]$ code plus a few extra words) produced nothing below 248; the best unions of cosets of a one-dimensional code have 250 words. Similarly, $K(19, 3) \le 1024$ and $K(21, 3) \le 3072$ coincide with $8K(13, 2)$ and $8K(15, 2)$, so improved normal codes for $K(13, 2)$ or $K(15, 2)$ would improve those entries as well. The density of our code, $1984 \cdot \sum_{i \le 3} \binom{20}{i} / 2^{20} \approx 2.56$, is still larger than that of the neighbouring entries (≈ 2.27 for $n = 19$ and ≈ 2.29 for $n = 21$), which suggests that $K(20, 3)$ can be improved further by a direct construction.

8 Limitations

The improvement is small and rests on a classical construction. Novelty is asserted only to our knowledge: Kéri's tables were last updated around 2009–2011 and an unpublished or obscure later improvement of $K(20, 3)$ cannot be completely excluded. The Lean proof certifies the theorem independently of the search code.

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